

Repair Approval Memo —

RAM-2025-002

CASCADIA RENEWABLE OPERATIONS LLC

Internal Memorandum — Operations, Engineering, and Procurement

Field	Value
Document ID	RAM-2025-002
Date	2025-09-18
From	Marcus Chen, Sr. Reliability Engineer · co-signed: Priya Ramachandran, Plant Manager, NW-PowerPool
To	Daniel Voss, Director of Operations · Elena Marquez, VP Engineering · cc: Lillian Park, CFO Office
Subject	ESCALATION — Decline \$1.2M Siemens Gamesa gearbox replacement (WIN-10131); authorize emergency stabilization only (EMR-2025-003, \$451K)
EPA Reference	EPA-2024-11 (invoked 2025-08-28)
Distribution	Operations Leadership · Engineering · CFO Office · Vendor Management

1. Background

On 2025-08-28, WIN-10131 (Siemens Gamesa wind turbine, NW-PowerPool-WIND-024) tripped on overheating at the main gearbox HSS bearing. The forced outage ran 190 hours and produced **4,632.9 MWh of lost generation** (outage OUT-2025-0570). CPS declined dispatch on crew availability; EPA-2024-11 was invoked and Apex Emergency Grid Solutions mobilized with Siemens Gamesa OEM engineering on-site.

The vendor reports (Apex **EMR-2025-003** and the on-site OEM concurrence) document micropitting plus a through-thickness crack on the HSS pinion. Apex completed stabilization and an auxiliary chiller install permitting derated operation. The vendor's emergency invoice is **\$451,000**.

This event follows two prior gearbox-class incidents on WIN-10131 in FY2025 (outages OUT-2025-0572 of 2025-01-14 and OUT-2025-0569 of 2025-03-20). WIN-10131 is on the FY2025 problem-asset list (17,239 MWh lost YTD; see RCA-2025-002).

2. Vendor Recommendation

Apex and Siemens Gamesa OEM engineering jointly recommend **full Siemens Gamesa OEM-supervised gearbox replacement** at WIN-10131, **\$1.2M per unit** (Siemens Gamesa quote **SGRE-Q-2025-1184**). The recommendation cites Siemens Gamesa OEM Manual Rev. 4.2 §8.3, which classifies the observed condition as Category I and prescribes gearbox replacement under OEM supervision. The OEM position is that field refurbishment voids the residual drivetrain warranty.

3. Cascadia Position

Cascadia Engineering does **not concur** with the OEM-mandated gearbox replacement at this time, on the following grounds:

1. **Cascadia Standard MS-2025-03** (issued February 2025) establishes vibration and wear-particle thresholds for refurbishment-versus-replacement determination on Siemens Gamesa gearboxes in our NW-PowerPool fleet. The MS-2025-03 thresholds are approximately 30% higher than the OEM Manual Rev. 4.2 §8.3 thresholds — meaning that the OEM manual triggers a replacement decision at conditions Cascadia engineering believes are appropriate for refurbishment.
2. **MOC-2025-022** (Management of Change request, filed March 2025) is the open request to formalize the MS-2025-03 override of OEM Manual Rev. 4.2 §8.3 for Cascadia operations. MOC-2025-022 remains in engineering review and is not yet adopted. We acknowledge the procedural gap.
3. The observed condition at WIN-10131 — micropitting and a 4.1 mm crack on tooth #14 — is, in our reading of MS-2025-03, refurbishment-eligible if executed under controlled conditions with an OEM-recognized refurbisher. We are also aware that Apex's pattern across FY2025 is to recommend replacement in every emergency event, which weighs in our evaluation.
4. The cumulative spend trajectory on WIN-10131 (3 events FY25, ~\$890K Apex invoicing) does not yet exceed the \$1.2M replacement cost. We have decision room to evaluate refurbishment.

4. Financial Justification

Line	Amount
Apex emergency stabilization (EMR-2025-003)	\$451,000
Siemens Gamesa gearbox replacement (declined)	\$1,200,000
Refurbishment alternative (estimated, separate quote pending)	\$380K–\$540K
Asset lost-generation YTD revenue impact (WIN-10131)	~\$1.03M at NW-PowerPool realized pricing

NW-PowerPool FY25 maintenance budget impact: approving the \$451K stabilization absorbs ~3.4% of the \$13.4M annual line. Declining the \$1.2M replacement defers a capital decision pending engineering closure on MS-2025-03 and MOC-2025-022.

5. Approval Request / Decision

Authorize the emergency stabilization invoice EMR-2025-003 at \$451K under EPA-2024-11. Do NOT authorize the Siemens Gamesa gearbox replacement at \$1.2M. Escalate the replacement decision to the Engineering steering committee with the following actions:

1. Engineering to issue a decision on **MOC-2025-022** within 60 days. The 9-month review timeline is itself a documented procedural risk (refer CAR-2024-118).
2. Reliability Engineering to procure an independent refurbishment quote from a non-OEM Siemens Gamesa-certified refurbisher, with timing to align with the MOC-2025-022 decision.
3. The auxiliary chiller skid (lease per EMR-2025-003) remains in service at WIN-10131 until the replacement-vs-refurbishment decision is final. Lease cost tracked under the same WO.
4. Cross-reference and elevate this case to **RCA-2025-002** (Fall Overheating Sequence root cause) and CAR-2025-002.

Approver	Role	Decision	Date
Marcus Chen	Sr. Reliability Engineer (author)	Approve stabilization; escalate replacement	2025-09-18
Priya Ramachandran	Plant Manager, NW-PowerPool	Concur	2025-09-18
Elena Marquez	VP Engineering		2025-09-19

Approver	Role	Decision	Date
		Concur — engineering will own MOC-2025-022 closure	
Daniel Voss	Director of Operations	Approve as escalated	2025-09-19
Lillian Park	CFO Office	Approve; flag \$1.2M conditional commitment in capital plan	2025-09-19

References: EMR-2025-003 · MS-2025-03 · Siemens Gamesa OEM Manual Rev. 4.2 §8.3 · MOC-2025-022 · CAR-2024-118 · CAR-2025-002 · RCA-2025-002 · OIR-2025-001 · EPA-2024-11.